

# Tests & Quizzes

## Quiz 1

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### Part 1 of 1 4.0 / 5.0 Points

Question 1 of 5

1.0

1.0 Points

#### **What is Bitcoin halving?**

- ☒ A. A pre-programmed event in which the block reward given to miners is reduced by 50%, occurring approximately every 210,000 blocks.
- ☐ B. A process where the total supply of Bitcoin is cut in half every four years to control inflation.
- ☐ C. A consensus mechanism change that reduces mining difficulty by 50% to maintain block time.
- ☐ D. A network upgrade that halves transaction fees to increase user adoption and scalability.

**Answer Key:** A

Question 2 of 5

0.0

1.0 Points

**Can tampering be detected in the latest block while it is still being mined?**

- ✓ A. No, because the block hash is not yet finalized during nonce computation.
- ✓ B. Yes, because nodes continuously verify block contents during mining.
- ✓ C. No, because hash linkage only applies after block confirmation.
- ✗ D. Yes, because every data change immediately invalidates the block hash.

**Answer Key:** A

Question 3 of 5

1.0

1.0 Points

**What is the purpose of the nonce in a blockchain block?**

- ✓ A. To uniquely label a block for identification by network nodes.
- ✓ B. To record the time at which the block is broadcast to the network.
- ✓ C. To store a value that helps verify transactions within the block.
- ✓ D. To provide a changing input used during hashing of the block header.

**Answer Key:** D

Question 4 of 5

1.0

1.0 Points

**A cryptocurrency follows Bitcoin's original issuance rules. If 19.91 million coins are already in circulation and 781,564 blocks have been produced, what amount of new currency is currently created when a valid block is added?**

- ✓ A. 12.528 units
- ✓ B. 12.345 units

- ✓ C. 6.250 units
- ✓ D. 1.862 units

**Answer Key:** C

Question 5 of 5

1.0

1.0 Points

**Which property of cryptographic hash functions most directly enables blockchain immutability?**

- ✓ A. Deterministic output for identical inputs.
- ✓ B. Computational resistance to preimage attacks.
- ✓ C. Fixed-length output regardless of input size.
- ✓ D. Sensitivity of output to any input modification.

**Answer Key:** D